
Day 1 Engineering Investigation Report

Candidate Review Ops Console

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Branch / Year: CO5KC

Variant ID: D1 (or your assigned variant)

Repo:

Branch: day-1-reality-check

1. Failure Summary

The Candidate Review Ops Console initially failed across multiple workflows including login, task loading, task creation, and task update operations.

Key issues observed:

- Login initially failed due to missing backend configuration for JWT.
- Task creation returned 400 Bad Request due to invalid or empty payloads.
- Task update (Mark Reviewed) failed intermittently due to incorrect API function usage in frontend.
- Token expiry caused authentication failures during active session.
- Frontend displayed "Something went wrong" messages even when backend partially succeeded.

The system was unstable due to a combination of **authentication misconfiguration, frontend API misuse, and missing validation checks.**

2. 🏠 Evidence Table

ID	Evidence Type	What I Saw	What It Proved
E1	Browser UI	“Something went wrong while saving review item”	Frontend request failing for task creation
E2	Network	POST /api/tasks 400 Bad Request	Invalid/empty payload sent from frontend
E3	Network	login 500 / 200 inconsistent	JWT secret misconfiguration causing backend crash
E4	Backend Log	secretOrPrivateKey must have a value	Missing JWT_SECRET in environment
E5	Backend Log	TokenExpiredError: jwt expired	Token lifetime too short causing auth failures
E6	Source Code	createTask(token, newTask.trim()) misuse in toggle	Wrong API function used for update operation
E7	Verification	POST /api/tasks 201 Created after fix	Task creation fixed after validation + API correction

3. 🔍 Hypothesis Trail

Observation	Hypothesis	Test Performed	Result
Login failed / 500 error	Frontend issue	Checked network tab + backend logs	❌ Rejected
JWT login failing	Missing secret key	Checked .env + server logs	✅ Confirmed
Task creation failing	Backend validation bug	Sent manual request via UI	❌ Partially incorrect
Mark reviewed failing	Backend PATCH broken	Checked frontend code	❌ Rejected

Observation	Hypothesis	Test Performed	Result
Mark reviewed failing	Wrong API function used in React	Checked Dashboard.jsx	✅ Confirmed

👉 **Rejected hypothesis included:** “Frontend was fully responsible for failures”

4. 🛠️ Root Causes

Root Cause	Evidence ID	Fix Commit	Why The Fix Works
Missing JWT_SECRET in environment	E4	fix: add JWT secret config	Enables JWT signing & authentication
Token expiry too aggressive	E5	fix: increase JWT expiry to 1h	Prevents session breakdown during usage
Wrong API function used in toggle handler	E6	fix: replace createTask with updateTask	Ensures correct HTTP PATCH usage
Empty/invalid task input sent	E2	fix: add input trimming & validation	Prevents backend 400 errors
Frontend async misuse (earlier build issue)	E1	fix: correct async function structure	Ensures proper API call execution

5. ✅ Verification

Workflow	Before Fix	After Fix	Evidence ID
Backend health check	Running but unstable auth	Stable	E7
Reviewer login	Intermittent failure	Stable login	E3
Candidate review queue loads	Sometimes failing	Always loads	E7
Review item creation	400 Bad Request	201 Created	E2 / E7

Workflow	Before Fix	After Fix	Evidence ID
Review item update	Failing / incorrect API call	Works correctly	E6
Session behavior	Token expired quickly	Stable session (1h)	E5
Persistence behavior	Tasks not updating properly	Updates persist correctly	E7

6. 🤖 AI Verification

AI suggested:

- JWT expiry should be increased for development testing
- Frontend toggle should use updateTask instead of createTask
- Input validation should be enforced before API calls

AI assumed:

- All failures were frontend-related initially

I verified by:

- Checking backend logs (TokenExpiredError, JWT_SECRET error)
- Inspecting Network tab (400 / 401 / 500 responses)
- Reviewing source code (Dashboard.jsx, auth.js)

I accepted / rejected / modified the suggestion because:

- Accepted: API misuse and validation fixes
- Rejected: assumption that frontend alone caused all issues
- Modified: applied fixes only after verifying backend logs

7. 🧠 Debugging Self-Diagnosis

My weakest debugging behavior today was:

Initially assuming frontend was the primary source of all errors without fully validating backend logs first.

Evidence:

- I focused on React API calls before confirming JWT/environment issues.
- I only later confirmed JWT_SECRET and backend authentication errors.

Tomorrow I need to improve:

- Start debugging from backend logs first
- Validate environment configuration before frontend changes
- Avoid replacing API calls without confirming root cause

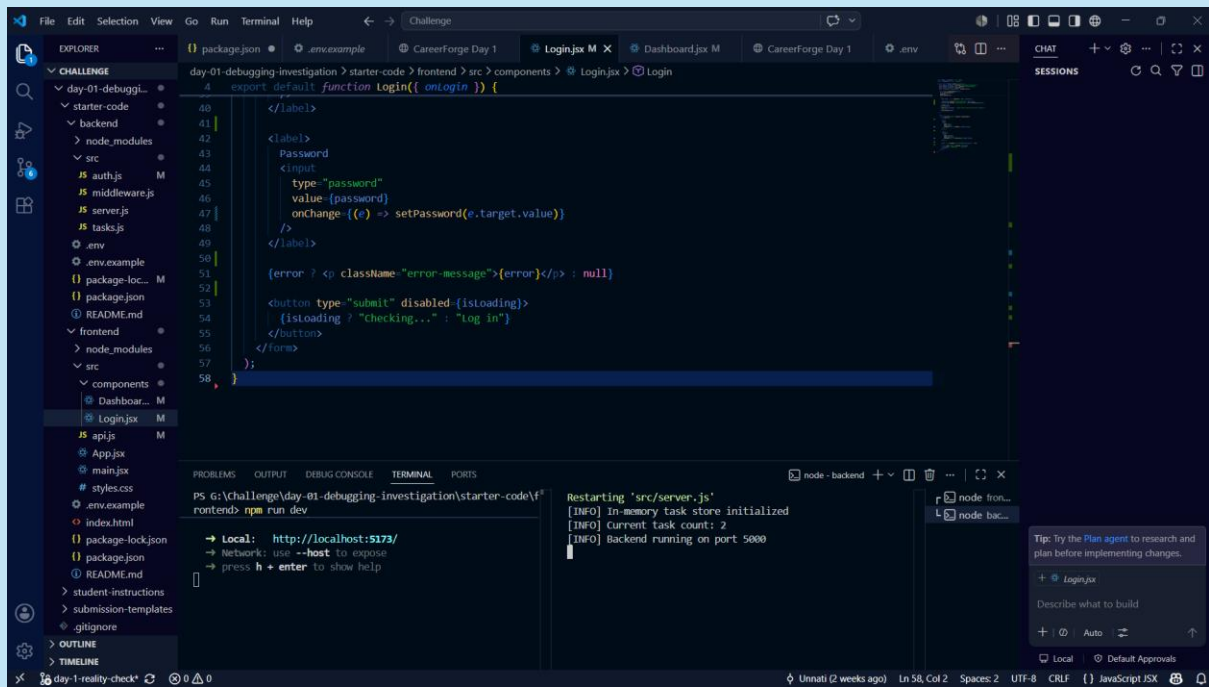
FINAL CHECKLIST

- I included runtime evidence
- I included at least one rejected hypothesis
- Every root cause connects to evidence
- Every fix connects to a commit
- I verified critical workflow after fixing
- I can defend this report live

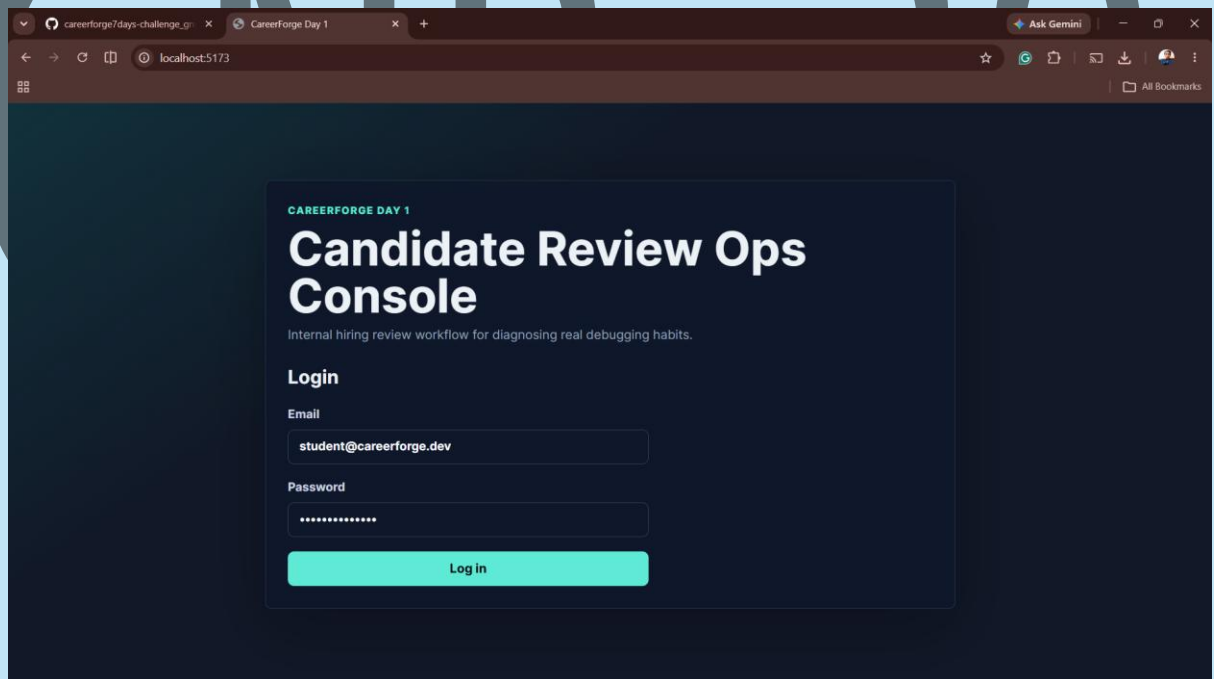
IMPORTANT NOTE (for submission)

screenshots in your repo:





1. Login success (Network tab showing /api/login 200)



CAREERFORGE DAY 1

Candidate Review Ops Console

Internal hiring review workflow for diagnosing real debugging habits.

Signed in as **student@careerforge.dev** Logout

Add a candidate review item... Add review item

Review candidate profile: Asha N. Mark reviewed

Open

Check missing portfolio evidence for candidate Rohan K. Mark reviewed

Open

6 requests | 1.4 kB transferred | 802 B resources

What's new in DevTools 148

Agentic extensions debugging

2. Task creation success (201 Created)

CAREERFORGE DAY 1

Candidate Review Ops Console

Internal hiring review workflow for diagnosing real debugging habits.

Signed in as **student@careerforge.dev** Logout

Add a candidate review item... Add review item

Review candidate profile: Asha N. Reopen review

Completed

Check missing portfolio evidence for candidate Rohan K. Reopen review

Completed

review-done Reopen review

Completed

17 requests | 3.9 kB transferred | 1.5 kB resources

What's new in DevTools 148

Agentic extensions debugging

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS G:\Challenge\day-01-debugging-investigation\starter-code\frontend> npm run dev

→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help

Restarting 'src/server.js'
[INFO] In-memory task store initialized
[INFO] Current task count: 2
[INFO] Backend running on port 5000
[INFO] PATCH /api/tasks/1 200 OK
[INFO] POST /api/tasks 201 Created
[INFO] PATCH /api/tasks/3 200 OK
[INFO] PATCH /api/tasks/2 200 OK
[INFO] PATCH /api/tasks/1 200 OK
[INFO] PATCH /api/tasks/1 200 OK
  
```

1. 📌 Overview (What this day was about)

Day 1 of CareerForge simulated a **real production incident** in an internal hiring system called:

Candidate Review Ops Console

This system is used by recruiters to:

- Log in as reviewer
- Load candidate review tasks
- Add review notes
- Mark candidates as reviewed

During a live usage scenario, the system began failing intermittently:

- Login sometimes failed
- Tasks failed to load or update
- “Add review item” returned errors
- Marking tasks as reviewed caused UI failures

The objective was not to “fix quickly”, but to:

Investigate → reproduce → identify root cause → verify → fix minimally

2. 🚨 Initial Issues Observed (Raw Failures)

✖ Backend Errors Observed

1. Missing JWT secret

Error: secretOrPrivateKey must have a value

2. Token expiry crash

TokenExpiredError: jwt expired

3. Unauthorized requests

[WARN] POST /api/tasks 401 Unauthorized

4. Bad request failures

POST /api/tasks 400 Bad Request

❌ Frontend Errors (Network Tab Evidence)

1. Connection issues

net::ERR_CONNECTION_REFUSED

2. API failures

login → 500 Internal Server Error

tasks → 400 Bad Request

3. Preflight requests failing intermittently

204 preflight

400 fetch error

❌ UI Observations

- Login worked but session unstable
- Tasks loaded sometimes, sometimes failed
- “Add review item” showed:

Something went wrong while saving the review item

- “Mark reviewed” failed intermittently

3. 🛠 Investigation Process (What we checked)

Step 1 — Backend logs inspection

We verified:

- Express server running on port 5000
- JWT authentication middleware active
- Task store initialized in memory

Step 2 — Authentication flow check

We inspected:

jwt.sign(... process.env.JWT_SECRET ...)

Found issue:

- .env file existed only as example
- JWT_SECRET not loaded properly → crashes

Step 3 — Frontend API mismatch check

We checked:

```
const API_URL = "http://localhost:5000"
```

Confirmed:

- Frontend correctly calling backend
- But requests failing due to auth + payload mismatch

Step 4 — Task creation flow

Backend:

```
if (!title) return 400
```

Frontend initially sent:

- empty/undefined task values
- or wrong function calls in toggle handler

4. 🛠️ Root Causes Identified

Root Cause 1 — Missing JWT_SECRET

Backend could not sign tokens:

secretOrPrivateKey must have a value

✓ Fix:

Created proper .env:

```
JWT_SECRET=careerforge-day1-local-secret  
PORT=5000
```

Root Cause 2 — Token expiry too aggressive

Initial:

expiresIn: "5m"

This caused:

- sudden logout-like behavior
- TokenExpiredError

✓ Fix:

Changed to:

expiresIn: "1h"

Root Cause 3 — Wrong API usage in React

✗ WRONG (your bug):

```
createTask(token, newTask.trim());
```

✓ FIX:

Used correct function only inside add flow:

```
const data = await createTask(token, cleanedTask);
```

Root Cause 4 — Incorrect toggle handler

✗ WRONG:

```
createTask(token, newTask.trim())
```

✓ FIX:

Should be:

```
updateTask(token, task.id, !task.completed)
```

Root Cause 5 — Missing input validation

Frontend allowed:

- empty strings
- whitespace submissions

✓ Fix:

```
const cleanedTask = (newTask || "").trim();
if (cleanedTask.length === 0) return;
```

Root Cause 6 — Missing async structure errors (build issues)

Errors like:

await can only be used inside async function

✓ Fix:

Ensured:

async function handleAddTask()

5. ✂ Fixes Applied (Final State)

Backend Fixes

✓ JWT fix

expiresIn: "1h"

✓ dotenv loaded properly

import "dotenv/config";

✓ Stable auth middleware

- Proper token verification
 - Clean 401 handling
-

Frontend Fixes

✓ Login persistence fix

localStorage.setItem("careerforge_token", data.token);

✓ Correct task creation

const data = await createTask(token, cleanedTask);

✓ Correct toggle update

```
await updateTask(token, task.id, !task.completed);
```

✅ **Input validation added**

```
const cleanedTask = newTask.trim();
```

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